# Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

Surnella Sandeep

Department of Data Analytics, Walsh College

QM640: Data Analytics Capstone

Faculty Mentor: Mr. Smitraj Raut

Interim Capstone Report (Ungraded Review Milestone)

October 20, 2026

**Project Repository & Data Availability:** https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

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# Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

## 1.1 Background and Problem Domain

Conversational artificial intelligence (AI) agents are increasingly used in travel and tourism to support destination discovery, itinerary planning, and booking (Jannach et al., 2021). Travelers have different needs and interaction styles. First-time travelers often seek reassurance and step-by-step guidance, solo travelers value flexibility, families prioritize comfort and accessibility, and business travelers focus on speed and efficiency (Inaba et al., 2022). However, most current travel chatbots rely on static, "one-size-fits-all" prompts. These rigid prompts treat every user identically and fail to adapt their questions or pacing to the traveler's specific needs. As a result, users face repetitive clarification questions and conversational friction, often abandoning the session before completing their plans.

To solve this issue, system designers often try to adapt the chatbot's style dynamically through prompt engineering, such as adding example dialogues into the prompt. However, adjusting conversation styles through prompts alone creates serious challenges (Rebedea et al., 2023). When unconstrained conversation examples are added to the prompt, large language models frequently get confused. They tend to copy outdated information from the examples, such as old ticket prices, invalid travel dates, or conflicting customer preferences. Worse, the model can hallucinate false details or violate strict travel business policies, such as booking without required passport validity or offering unauthorized discounts. To address these challenges, this research develops a data-driven framework that groups travel conversations into distinct styles to guide the prompt dynamically, while using strict business rules to prevent factual errors and policy violations. This approach aims to provide personalized, style-appropriate conversations while keeping travel bookings accurate and compliant.

### 1.2 Theoretical Foundation and Prior Art (MultiWOZ 2.2, Frames, S-BERT)

This study builds directly upon foundational literature in dialogue state tracking, sentence representation learning, and in-context learning. The primary theoretical baseline is established by the MultiWOZ benchmark (Budzianowski et al., 2018), specifically the MultiWOZ 2.2 release (Zang et al., 2020), which standardized dialogue act schemas, cleaned state annotation errors, and introduced schema-guided dialogue ontology definitions. To model multi-intent, negotiation-heavy travel interactions where customers dynamically revise constraints across turns, this study integrates the Microsoft Frames corpus (El Asri et al., 2017). Frames formalizes frame tracking, where conversational agents maintain multiple competing candidate itineraries as users explore trade-offs between flight dates, budget limits, and cabin classes.

In the representation domain, we leverage Sentence-BERT (Reimers & Gurevych, 2019) to construct dense semantic trajectory embeddings. Whereas traditional dialogue analysis relies on bag-of-words or turn-level surface lexical matching, S-BERT utilizes siamese and triplet networks to map entire conversational turn sequences into semantically meaningful vector spaces where geometric distance reflects structural conversational behavior. Finally, this research extends the in-context few-shot learning paradigm formalized by Brown et al. (2020) and refined by Liu et al. (2022). Liu et al. demonstrated that LLM downstream task performance is heavily governed by the semantic proximity of retrieved in-context exemplars. We advance this theory by demonstrating that in-context exemplars act as powerful stylistic conditioning priors that steer conversational tone, pacing, and turn economy without requiring parameter fine-tuning.

### 1.3 Problem Statement (Independent and Dependent Variables)

Existing goal-oriented conversational systems lack data-driven mechanisms to automatically discover latent user conversational archetypes from historical dialogue logs and

adapt their interaction style in real-time, resulting in conversational friction, redundant elicitation, and high abandonment. Furthermore, unconstrained prompt-based style steering introduces factual leakage and policy drift that commercial platforms cannot tolerate.

In a quantitative data analytics framework, this research evaluates how key independent variables like Clustering Algorithm, In-Context Conditioning Strategy, Business Rule Enforcement Mode, and In-Context Exemplar Count, statistically impact measurable dependent variables like Structural Cluster Quality, Semantic Style Convergence, Factual Slot Accuracy, Exemplar Factual Leakage Rate, Operational Rule Adherence Rate, and Conversational Efficiency (Detailed operational definitions, metric scales, and analytical roles for each variable are formalized in Table 2).

### 1.4 Purpose of the Study

The purpose of this quantitative data analytics study is fourfold: (a) to discover, evaluate, and interpret latent conversational archetypes from human-human travel dialogues using dense trajectory embeddings; (b) to quantify the degree of semantic and stylistic convergence achieved by dynamic cluster-adaptive few-shot prompt injection compared to static baselines; (c) to conduct a controlled 2x2 ablation experiment isolating the causal impact of declarative business rules versus few-shot conditioning on factual slot preservation and policy adherence; and (d) to evaluate conversational efficiency gains across distinct traveler archetypes. The investigation delivers a rigorous, fully reproducible methodology for context-adaptive conversational AI.

### 1.5 Scope and Analytical Boundaries

The scope of this research is strictly bounded to open-access, multi-turn English-language goal-oriented dialogue benchmark datasets, specifically the Microsoft Frames corpus (1,369 dialogues) and the MultiWOZ 2.2 benchmark (10,438 dialogues). The investigation utilizes

dense sentence representation models (Sentence-BERT all-MiniLM-L6-v2) to vectorize conversational turn trajectories and evaluates distance- and density-based clustering algorithms operating in Euclidean and cosine metric spaces. By relying exclusively on published, peer-reviewed public data, the methodology guarantees 100% scientific reproducibility without proprietary data dependencies

### 1.6 Research Questions (RQ1–RQ4)

To guide the investigation, four focused research questions are established:

• Research Question 1 (RQ1 - Unsupervised Trajectory Clustering Separation): Which unsupervised clustering algorithm (K-Means, Agglomerative Hierarchical, or HDBSCAN) operating on dense trajectory embeddings produces the highest structural cluster separation (Silhouette Score and Davies-Bouldin Index) and distinct behavioral archetypes across multi-turn goal-oriented dialogues?

• Research Question 2 (RQ2 - Dynamic In-Context Semantic Style Convergence): To what extent does dynamic cluster-adaptive few-shot exemplar injection increase the agent's Semantic Style Closeness (Cosine Similarity and BERTScore to target reference cluster centroids) compared to zero-shot and static few-shot baselines?

• Research Question 3 (RQ3 - Controlled Ablation: Rules vs. Style Factual Drift): In a controlled 2x2 ablation experiment (Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]), does aggressive stylistic convergence induce Exemplar Factual Leakage or degrade Slot F1, and does a decoupled declarative rule engine eliminate policy violations without dampening style convergence?

• Research Question 4 (RQ4 - Turn Economy & Friction Across Archetypes): How does cluster-adaptive few-shot prompting impact conversational efficiency, measured via Turns to

Goal Completion (TGC) and Redundant Elicitation Rate (RER), when interacting across distinct customer conversational archetypes (e.g., Consultative/Ambiguous vs. Direct/Transactional)?

## 2. Research Methodology, Randomization, and Sample Size Determination

### 2.1 Method Selection, Golden Partitioning, and Power Rationale

To ensure robust statistical validity and prevent Type I and Type II errors, sample size determinations were established across each research question. One-Way Analysis of Variance (ANOVA) power analysis was selected for RQ1 and RQ2 to test differences across clustering algorithms and prompt conditioning treatments. A Two-Way Factorial ANOVA power calculation was applied to the 2x2 ablation design in RQ3. For RQ4, the Confidence Interval method for population means was applied to estimate dialogue efficiency metrics within a tight margin of error. Standard statistical criteria are enforced across all evaluations: significance level alpha = .05 (two-tailed, $Z_{\alpha/2} = 1.96$) and statistical power 1 - beta = .80 ($Z_{\beta} = 0.8416$).

Train-Test Separation and Golden Reference Architecture: In quantitative machine learning and conversational analytics, strict separation between training and evaluation partitions is essential to eliminate data contamination (data leakage). The primary Microsoft Frames benchmark (1,369 dialogues) is formally partitioned into two disjoint subsets:

1. Golden Reference and Retrieval Pool ($D_{gold}$, 70%, $N_{train}$ approx 958 dialogues): High-performing Wizard-of-Oz conversations filtered for complete task success (saturated Tier-1 mandatory slots, verified booking commitment, and zero mid-session abandonment) operationalize the empirical gold standard. This subset is used exclusively to train unsupervised clustering algorithms, establish canonical archetype centroids $\{C_1, C_2, C_3\}$, and populate the FAISS vector index for few-shot exemplar retrieval. Crucially, the frozen centroids establish

fixed reference coordinates against which trajectory deviation and semantic style convergence are subsequently evaluated.

2. Held-Out Evaluation Test Cohort (D_test, 30%, N = 390 dialogues): In accordance with machine learning validation standards, this evaluation cohort remains completely unobserved during training, clustering, and centroid derivation (D_gold intersect D_test = empty set). All evaluation experiments across RQ1–RQ4 execute strictly on this test partition, satisfying the minimax sample size requirement derived in Section 2.3. When the conversational agent interacts with unseen test scenarios, the dynamic prompt assembler retrieves k = 3 stylistic exemplars from D_gold without exposing test labels or full dialogue trajectories, guaranteeing zero data leakage.

Randomization and Bias Mitigation Strategy: To prevent selection, temporal, and ordering artifacts, dialogue instances across all corpora are subjected to uniform random shuffling prior to experimental partitioning. Cross-validation splits are generated using Stratified ShuffleSplit across 10 repeated iterations with fixed pseudorandom seeds, guaranteeing balanced archetype representation and eliminating ordering artifacts.

## 2.2 Stepwise Sample Size Calculations (RQ1–RQ4)

### *2.2.1 Stepwise Calculation for RQ1: One-Way ANOVA (Cluster Quality Differences)*

RQ1 compares Silhouette Scores across three unsupervised clustering algorithms (K-Means, agglomerative hierarchical clustering, and HDBSCAN). The calculation assumes a medium standardized effect size of Cohen's f = 0.25, a significance level of alpha = .05, statistical power of 1 - beta = .80, and k = 3 comparison groups. With df_groups = k - 1 = 2, critical F_(.05, 2, inf) = 3.00, and noncentrality parameter lambda = 9.64, the sample size derivation proceeds across four sequential steps:

**Step 1. Squared effect size calculation:** f^2 = (0.25)^2 = 0.0625.

**Step 2. Omnibus total sample size calculation:** Using Cohen's noncentrality formula, N_total = lambda / f^2 = 9.64 / 0.0625 = 154.24 dialogues across all three algorithms combined.

**Step 3. Per-group allocation calculation:** With k = 3 groups, the per-algorithm requirement is n = ceil(154.24 / 3) = ceil(51.41) = 52 dialogues per clustering model.

**Step 4. Final total sample size determination:** Across the three algorithms, the final target sample size is N_RQ1 = 3 * 52 = 156 cluster evaluation units.

### *2.2.2 Stepwise Calculation for RQ2: One-Way ANOVA (Semantic Style Convergence)*

RQ2 assesses semantic cosine similarity and BERTScore across three prompt conditioning arms: Zero-Shot, Static Few-Shot, and Cluster-Adaptive Few-Shot. The calculation specifies an anticipated medium effect size of Cohen's f = 0.28, alpha = .05, statistical power of 1 - beta = .80, and k = 3 treatment groups (lambda = 9.64):

**Step 1. Squared effect size calculation:** f^2 = (0.28)^2 = 0.0784.

**Step 2. Omnibus total sample size calculation:** N_total = lambda / f^2 = 9.64 / 0.0784 = 122.96 dialogues across all three conditioning arms combined.

**Step 3. Per-group allocation calculation:** With k = 3 arms, the baseline allocation is n = ceil(122.96 / 3) = ceil(40.98) = 41 dialogues per arm. Rounding to maintain balanced cross-validation partitions yields n = 42 dialogues per arm.

**Step 4. Final total sample size determination:** Across the three prompt conditioning strategies, the final target sample size is N_RQ2 = 3 * 42 = 126 dialogues.

### *2.2.3 Stepwise Calculation for RQ3: Two-Way Factorial ANOVA (2x2 Ablation Experiment)*

RQ3 evaluates the 2x2 factorial ablation design (Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]) on Factual Slot F1 and Operational Rule Adherence Rate. For a four-cell factorial design, detecting a main effect or interaction effect of size f = 0.25 at alpha = .05 and power = .80 requires numerator df = 1, critical F_(.05, 1, inf) = 3.84, and noncentrality lambda = 7.85:

**Step 1. Baseline total sample size calculation:** N_base = lambda / f^2 = 7.85 / 0.0625 = 125.60 dialogues.

**Step 2. Operational buffer adjustment:** Incorporating a 40% contingency buffer to protect against incomplete dialogue runs or synthetic generation failures yields N_buffered = 125.60 * 1.40 = 175.84 dialogues.

**Step 3. Per-cell allocation calculation:** Allocating equally across the four factorial cells yields n = ceil(175.84 / 4) = ceil(43.96) = 45 dialogues per experimental cell.

**Step 4. Final total sample size determination:** Across the four factorial treatment arms, the final target sample size is N_RQ3 = 4 * 45 = 180 multi-turn dialogues.

### *2.2.4 Stepwise Calculation for RQ4: Confidence Interval Estimation (Turn Economy and Friction)*

RQ4 estimates population mean Turns to Goal Completion (TGC) and Redundant Elicitation Rate (RER) within a precision margin of error E = 0.35 turns (representing 10% of standard deviation sigma = 3.5 turns; El Asri et al., 2017) at a 95% confidence level (two-tailed Z_alpha/2 = 1.96):

**Step 1. Squared critical product calculation:** (Z_alpha/2 * sigma)^2 = (1.96 * 3.5)^2 = (6.86)^2 = 47.0596.

**Step 2. Squared margin of error calculation:** E^2 = (0.35)^2 = 0.1225.

**Step 3. Unrounded sample size calculation:** N_raw = 47.0596 / 0.1225 = 384.16 dialogues.

**Step 4. Final total sample size determination:** Rounding up to the next integer yields N_RQ4 = 385 dialogue interactions evaluated across customer archetypes.

### 2.3 Sample Size Summary and Minimax Decision (N = 390)

To ensure all four research questions achieve full statistical validity without under-powering any subgroup analysis, the final sample size decision follows the formal minimax decision rule: N_final = max(N_RQ1, N_RQ2, N_RQ3, N_RQ4) = max(156, 126, 180, 385) = 385 dialogues. To facilitate balanced cross-validation across 3 cluster archetypes and 4 experimental folds, the sample size is rounded to exactly N = 390 dialogues (providing 130 dialogues per behavioral archetype). This cohort constitutes the 30% held-out test split drawn from Microsoft Frames, guaranteeing full statistical power using public data alone. This final decision is highlighted in Table 1 below.

**Table 1**

*Sample Size Determination Summary and Final Minimax Decision across Research Questions*

| RQ # | Analytical Focus | Method | Alpha | Power | Target f / d | Required N |
|---|---|---|---|---|---|---|
| RQ1 | Cluster Separation (Silhouette) | One-Way ANOVA | 0.05 | 0.80 | f = 0.25 | N1 = 156 |
| RQ2 | Style Convergence (Cosine Sim) | One-Way ANOVA | 0.05 | 0.80 | f = 0.28 | N2 = 126 |
| RQ3 | 2x2 Ablation (Rules vs Prompting) | 2x2 Fact. ANOVA | 0.05 | 0.80 | f = 0.25 | N3 = 180 |
| RQ4 | Turn Economy (Turns to Goal) | Confidence Int. | 0.05 | N/A | E = 0.35, s=3.5 | N4 = 385 |
| **Final** | **Comprehensive Study Cohort** | **Minimax Decision** | **0.05** | **>= 0.80** | **Max(RQ1-RQ4)** | **N = 390** |

*Note. Final sample size N = 390 satisfies the minimax criterion N = max(N_RQ1..N_RQ4) and enables equal partition across 3 cluster archetypes.*

## 3. Data Infrastructure, Two-Tier Ontology, and Project Repository

### 3.1 Primary Open Research Datasets and Access Links

To guarantee 100% academic reproducibility, this research utilizes widely recognized, open-access multi-turn conversational corpora. The primary training and clustering dataset is the Microsoft Frames corpus (El Asri et al., 2017), comprising 1,369 human-human dialogues (19,986 individual turns) collected through a Wizard-of-Oz interface. In Frames, users search for vacation packages while exploring complex constraints (dates, destination trade-offs, budget adjustments, and cabin classes). The corpus is publicly accessible at: https://www.microsoft.com/en-us/research/project/frames-dataset/

For schema validation and dialogue state tracking cross-evaluation, we incorporate MultiWOZ 2.2 (Zang et al., 2020), which provides 10,438 multi-domain dialogues spanning hotel, attraction, flight, and taxi domains. MultiWOZ 2.2 features standardized slot definitions and corrected dialogue state labels, accessible at: https://github.com/budzianowski/multiwoz

### 3.2 Two-Tiered Canonical Slot Schema (Meta-Ontology Alignment)

A central methodological challenge when combining diverse conversational datasets is ontology divergence. Microsoft Frames uses slot labels such as or_city, dst_city, str_date, and budget, whereas MultiWOZ employs hotel-pricerange, flight-arriveBy, and taxi-departure. Furthermore, operational travel agents require policy constraints (e.g., passport validity, advance booking windows) absent from raw conversational text. To resolve this, we adopt the meta-ontology alignment methodology formalized in MultiWOZ 2.2 (Zang et al., 2020) and construct a Two-Tier Canonical Slot Schema:

**Tier 1 (Universal Core Slots):** Standardized across all datasets and models, including: Origin_City, Destination_City, Departure_Date, Return_Date, Duration_Nights, Traveler_Count (Adults/Children), Budget_Amount, Accommodation_Tier, and Booking_Status.

**Tier 2 (Operational Extension Slots):** Policy-level operational parameters evaluated in RQ3/RQ4, including: Visa_Processing_Window_Days, Passport_Expiry_Months, Package_Type (Group/FIT), Sharing_Basis (Twin/Single), and Discount_Applied_Percentage.

This two-tier schema ensures seamless slot mapping across heterogeneous datasets while isolating operational policy rules from prompt formatting.

### 3.3 Cross-Corpus Reference Clustering: Frames and MultiWOZ Transfer

To validate whether conversational trajectory archetypes discovered in Microsoft Frames generalize across independent data distributions, we conduct cross-corpus domain adaptation against MultiWOZ 2.2 (travel and accommodation domains). Because Frames focuses on multi-turn constraint trade-offs whereas MultiWOZ models slot-directed task completion, this cross-corpus evaluation tests the structural stability of the learned latent space. To prevent spurious lexical clustering driven by entity tokens (specific hotel or city names), all named entities and currency expressions are normalized to generic tokens ([DESTINATION], [ORIGIN], [HOTEL_NAME], [CURRENCY]) prior to vectorization. Dialogues from MultiWOZ 2.2 are projected directly into the latent trajectory space established by Frames, evaluating whether the discovered behavioral archetypes (Hesitant Explorer, Budget Negotiator, Rapid Transactional) demonstrate geometric clustering invariance across distinct open-access corpora.

### 3.4 Ethical Considerations, Bias Mitigation, and Random Shuffling

Because conversational AI agents can inadvertently amplify demographic stereotypes or exhibit exclusionary behavior, rigorous ethical safeguards are enforced. First, data privacy and intellectual property are fully protected: this research utilizes exclusively open-access, anonymized public corpora (Microsoft Frames and MultiWOZ 2.2) distributed under permissive research licenses, completely eliminating confidentiality, corporate non-disclosure, or personal data leakage risks. Second, algorithmic bias in few-shot conditioning is actively mitigated. Cluster archetypes are defined purely on conversational interaction mechanics (interaction length, turn latency, constraint revision frequency) rather than user demographic attributes,

ensuring equitable agent behavior. Finally, uniform random permutation of dialogue logs is enforced across training splits to ensure cluster definitions remain invariant to arrival order.

### 3.5 Project GitHub Repository and Verification

To satisfy project reproducibility and open-science standards, a dedicated, accessible public GitHub repository has been established at: https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. The repository houses all data ingestion scripts, Pydantic schemas, clustering pipelines, FAISS retrieval modules, declarative rule configurations, and automated evaluation harnesses. The complete repository directory structure, component schema, and execution commands are documented in Appendix A.

## 4. Comprehensive Data Dictionary

Table 2 provides the complete data dictionary defining all experimental variables, operational measures, data types, units, and methodological roles for evaluating RQ1 through RQ4. The framework evaluates six core constructs: Structural Cluster Quality (cluster cohesion and separation), Semantic Style Convergence (archetype alignment), Factual Slot Accuracy (constraint precision), Exemplar Factual Leakage Rate (prompt-induced hallucination), Operational Rule Adherence Rate (business policy compliance), and Conversational Efficiency (turn economy and friction reduction).

**Table 2**

*Comprehensive Experimental Data Dictionary for Goal-Oriented Dialogue Clustering and Adaptive Prompting*

| Variable Name | Data Type | Role | Operational Definition & Meaning | Units / Range | Relevance |
|---|---|---|---|---|---|
| dialogue_id | String | Identifier | Unique dialogue session code from Frames/MultiWOZ | Discrete ID | Unit of analysis for trajectory tracking |

| | | | | | |
|---|---|---|---|---|---|
| clustering_algo | Categorical | Independent | Clustering Algorithm: Unsupervised clustering method (K-Means, Agglomerative, HDBSCAN) | 3 Categories | Primary factor for RQ1 cluster evaluation |
| conditioning_mode | Categorical | Independent | In-Context Conditioning Strategy: Prompting mode (Zero-Shot, Static Few-Shot, Cluster Few-Shot) | 3 Categories | Treatment factor for RQ2 style convergence |
| rule_engine_mode | Categorical | Independent | Business Rule Enforcement Mode: Ablation condition (0=Rules OFF, 1=Declarative Rules Active) | Binary {0, 1} | Treatment factor for RQ3 ablation experiment |
| target_archetype | Categorical | Independent | Target Behavioral Archetype: Assigned dialogue cluster (Explorer, Negotiator, Transactional) | 3 Clusters | Stratification factor across RQ2–RQ4 |
| exemplar_count_k | Integer | Independent | In-Context Exemplar Count: Number of demonstration dialogues injected into prompt context (k in {1, 3, 5}) | Count (k in {1, 3, 5}) | Evaluates demonstration density effects |
| silhouette_score | Float | Dependent | Structural Cluster Quality: Cluster cohesion and separation in trajectory space | Score [-1.0, 1.0] | Primary metric for RQ1 evaluation |
| davies_bouldin_idx | Float | Dependent | Structural Cluster Quality: Ratio of within-cluster dispersion to separation | Index [0.0, inf) | Validation metric for RQ1 evaluation |
| cosine_similarity | Float | Dependent | Semantic Style Convergence: Directional similarity to target cluster centroid | Score [-1.0, 1.0] | Primary style convergence metric for RQ2 |
| bert_score_f1 | Float | Dependent | Semantic Style Convergence: Token-level semantic similarity to reference exemplars | Score [0.0, 1.0] | Secondary semantic alignment metric for RQ2 |
| slot_f1_score | Float | Dependent | Factual Slot Accuracy: Harmonic mean of slot precision and recall against user constraints | Ratio [0.0, 1.0] | Factual slot preservation metric for RQ3 |
| factual_leakage_rate | Float | Dependent | Exemplar Factual Leakage Rate: Proportion of hallucinated entities copied from prompt exemplars | Ratio [0.0, 1.0] | Direct measure of prompt contamination in RQ3 |
| rule_adherence_rate | Float | Dependent | Operational Rule Adherence Rate: Proportion of turns adhering to business rules (passport, party limits, blackouts) | Ratio [0.0, 1.0] | Primary compliance metric for RQ3 ablation |
| turns_to_goal | Integer | Dependent | Conversational Efficiency: Total interaction turns required to resolve user travel goal | Turns (1–25) | Primary conversational efficiency metric |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | for RQ4 |
| redundant_elicit_rate | Float | Dependent | Conversational Efficiency: Proportion of turns re-asking established constraints (friction) | Ratio [0.0, 1.0] | Dialogue friction metric for RQ4 |
| budget_gap_closure | Float | Dependent | Commercial Negotiation: Proportion of initial budget discrepancy resolved | Ratio [0.0, 1.0] | Commercial negotiation metric for RQ4 |

*Note. Variables span trajectory embeddings, prompt conditioning modes, ablation indicators, and evaluation metrics.*

## 5. Analytic Approach and Solution to RQ1–RQ4

This section maps each of the four research questions to its rigorous analytic methodology, detailed technical feature mechanics, mathematical formulations, procedural steps, and expected empirical outputs. Underlying statistical and geometric assumptions are formalized in Appendix D.

### 5.1 Solution to RQ1: Unsupervised Trajectory Clustering Quality

#### *5.1.1 Trajectory Construction Mechanics and Mathematical Formulation*

A foundational contribution of this research is the transformation of multi-turn conversational interactions into dense structural trajectory vectors. Let a completed dialogue session D of length T turns be represented as an ordered sequence of user and assistant turn pairs: D = {(u_1, a_1), (u_2, a_2), ..., (u_T, a_T)}. Converting D into a fixed-length representation suitable for distance-based clustering requires capturing both semantic evolution and interaction structure across three sequential steps:

**Step 1. Dense semantic turn mapping:** Each individual turn pair t is mapped into a d = 384-dimensional dense semantic vector using Sentence-BERT (all-MiniLM-L6-v2): e_t = S-BERT(u_t || ' [SEP] ' || a_t) in R^384.

**Step 2. Exponential temporal decay weighting:** Because early turns typically define constraint discovery while terminal turns capture negotiation closure, an exponential decay weighting function w_t is applied across turns: w_t = exp(lambda * t / T) / sum_{k=1}^T exp(lambda * k / T), where lambda = 0.75 is the progression parameter governing trajectory sensitivity.

**Step 3. Auxiliary structural mechanics integration:** To capture non-lexical conversational mechanics (pacing, hesitation, negotiation friction), an auxiliary structural feature vector s_D in R^6 is computed: s_D = [ ln(T), L_user_avg, L_asst_avg, Delta_budget_mention, Turn_to_First_Offer, Constraint_Revision_Count ], where L represents character length and Delta indicates pricing adjustment count. The final unified trajectory vector v_D is constructed via concatenation and L2-normalization: v_D = L2_Normalize( [ sum_{t=1}^T w_t * e_t ] || StandardScaler(s_D) ) in R^390.

### *5.1.2 Hypotheses, Analytic Method, and Evaluation Outputs*

**Null Hypothesis (H0):** There is no significant difference in structural cluster separation (Silhouette Score) across K-Means, Agglomerative Hierarchical, and HDBSCAN algorithms (mu_K = mu_Agg = mu_HDB).

**Alternative Hypothesis (H1):** Significant differences exist in structural cluster separation across algorithms, with density-based or agglomerative methods outperforming standard spherical K-Means (mu_HDB != mu_K).

**Analytical Method:** A One-Way Analysis of Variance (ANOVA) followed by Tukey's Honestly Significant Difference (HSD) post-hoc test evaluates Silhouette Scores across 10 repeated 5-fold cross-validation bootstrap iterations (N = 156 evaluation units).

**Mathematical Formulations:** The Silhouette Score for trajectory i is: s(i) = (b(i) - a(i)) / max(a(i), b(i)), where a(i) is mean intra-cluster distance and b(i) is mean nearest-cluster distance. The Davies-Bouldin Index is: DB = (1/k) * sum_{i=1}^k max_{j != i} ( (sigma_i + sigma_j) / d(c_i, c_j) ).

**Expected Outputs:** Identification of optimal k = 3 behavioral archetypes (Hesitant Explorer, Budget Negotiator, Rapid Transactional) with Silhouette Score >= 0.42 and Davies-Bouldin Index <= 1.15.

## 5.2 Solution to RQ2: Dynamic In-Context Few-Shot Style Convergence

### *5.2.1 Exemplar Selection Logic, Centroid Triggers, and Vector Retrieval*

To operationalize adaptive prompt steering during live multi-turn interactions, a precise exemplar retrieval trigger and matching pipeline is established. Exemplar selection operates across four sequential steps:

**Step 1. Dynamic retrieval trigger execution:** The exemplar selection mechanism is triggered at session initialization (turn t = 1) using the user's opening intent, and is re-evaluated at turn t = 2 upon receipt of the first constraint preference expression. The cumulative ongoing dialogue prefix P_t = {(u_1, a_1), ..., u_t} is vectorized in real time: q_t = S-BERT( sum_{i=1}^t w_i * Turn_i ).

**Step 2. Centroid proximity matching:** Query vector q_t is compared against precomputed cluster centroids {C_1, C_2, ..., C_K} established on the Golden Training Pool D_gold using cosine

similarity: k* = argmax_{j in {1..K}} ( (q_t . C_j) / (||q_t|| ||C_j||) ). If max cosine similarity exceeds confidence threshold tau_conf >= 0.65, the interaction is assigned to target archetype k*.

**Step 3. FAISS sub-index nearest-neighbor query:** Within the isolated FAISS vector sub-index I_{k*}, an exact inner-product search retrieves the top k = 3 most structurally similar historical golden dialogues from D_gold: E = Top-3_{d in I_{k*}} ( (q_t . d) / (||q_t|| ||d||) ).

**Step 4. Slot-masked prompt formatting:** The retrieved golden dialogues undergo entity de-lexicalization (replacing specific dates, cities, and prices with canonical slot tokens) before being formatted into the Gemini Flash API system prompt prefix, ensuring the LLM absorbs the conversational style, tone, and pacing of archetype k* without contaminating dialogue state with stale entity values.

### *5.2.2 Hypotheses, Analytic Method, and Evaluation Outputs*

**Null Hypothesis (H0):** Dynamic cluster-adaptive few-shot conditioning does not significantly increase Semantic Cosine Similarity to the target reference cluster compared to static few-shot prompting (mu_cluster = mu_static).

**Alternative Hypothesis (H1):** Dynamic cluster-adaptive few-shot conditioning produces significantly higher Semantic Cosine Similarity and BERTScore to target reference cluster centroids (mu_cluster > mu_static).

**Analytical Method:** A One-Way Repeated Measures ANOVA compares Cosine Similarity across three prompting treatments (Zero-Shot Baseline, Static Few-Shot, Dynamic Cluster Few-Shot) evaluated across N = 126 dialogue sessions (42 per condition) drawn from the held-out test cohort D_test.

**Mathematical Formulations:** Cosine Similarity between generated system response trajectory g and target cluster centroid c is: Cos_Sim = (g . c) / (||g|| * ||c||). BERTScore computes token-level greedy matching cosine similarity using contextualized RoBERTa embeddings.

**Expected Outputs:** Demonstration of statistically significant style convergence (Cosine Similarity increase from 0.58 baseline to >= 0.82 under dynamic conditioning, $p < .001$, Cohen's d >= 0.85).

## 5.3 Solution to RQ3: Controlled 2x2 Factorial Ablation of Rules vs. Style Factual Drift

### *5.3.1 Deterministic Goal Completion Criteria and Multi-Factor Success Rule*

To rigorously quantify whether conversational style convergence introduces factual hallucination or compromises task success, a deterministic, unambiguous goal completion rule is enforced. A dialogue session D is classified as a binary success (Y_success = 1) if and only if three sequential criteria are simultaneously satisfied:

**Step 1. Mandatory slot saturation verification:** All Tier-1 mandatory core slots (Origin_City, Destination_City, Departure_Date, Duration_Nights, Traveler_Count, Budget_Tier) must be fully populated in the final dialogue state tracker: State_slots intersect Required_slots = Required_slots.

**Step 2. Operational policy constraint compliance:** Automated programmatic evaluation against the declarative rule schema yields zero blocking violations (V_blocking = 0), verifying that no package was offered violating visa lead times, exceeding customer budget caps, or breaching passenger capacity limits.

**Step 3. Explicit terminal commitment act execution:** The assistant successfully executes a formal commitment action (Offer_Package or Confirm_Itinerary) and the user responds with an explicit acceptance dialogue act (Inform(Accept) or Affirm()), concluding before max turns T <= 20: Success(D) = I( (Slot_Saturation == True) and (V_blocking == 0) and (Terminal_Acceptance == True) ).

### *5.3.2 Hypotheses, Analytic Method, and Evaluation Outputs*

**Null Hypothesis (H0):** Active declarative rules do not significantly improve Rule Adherence Rate (RAR) or prevent factual leakage under cluster conditioning (Main Effect beta_rules = 0).

**Alternative Hypothesis (H1):** Active declarative rules significantly increase Rule Adherence Rate (RAR >= 0.98) and eliminate factual leakage, proving business rules causally enforce policy compliance without dampening style convergence.

**Analytical Method:** A Two-Way Factorial ANOVA (2x2 design: Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]) evaluates Slot F1, Exemplar Leakage Rate, and Rule Adherence across N = 180 dialogues (45 per cell) drawn from held-out cohort D_test.

**Mathematical Formulations:** Exemplar Factual Leakage Rate is: Leakage = (Hallucinated Entities Traced to Prompt Exemplars) / (Total Generated Entities). Rule Adherence Rate is: RAR = 1 - (Violations / System Turns).

**Expected Outputs:** Proof that declarative rules eliminate factual leakage (reducing leakage from ~14.2% in Rules-OFF down to 0.0% in Rules-Active) while maintaining high Slot F1 (>= 0.92).

## 5.4 Solution to RQ4: Dialogue Turn Economy and Friction Analysis

### *5.4.1 Declarative Rule Schema Scope and Precise Boundary Conditions*

To establish reproducible boundaries for conversational efficiency and compliance, the declarative rule engine (rules_declarative.yaml) enforces four explicit classes of operational boundary conditions validated deterministically across four sequential steps:

**Step 1. Temporal and visa feasibility boundary enforcement:** Minimum booking lead time requires international departures >= 21 calendar days when visa processing is pending (domestic >= 48 hours), and international bookings require passport validity >= 6 months beyond the scheduled return date.

**Step 2. Financial and commercial boundary enforcement:** Maximum discount ceiling prohibits negotiated promotional discounts exceeding 10.0% of catalog base fare without manager escalation, and hard budget cap adherence strictly forbids proposing itineraries where Price > B_max.

**Step 3. Capacity and operational boundary enforcement:** Minor accompaniment requires child travelers (< 18 years) to have >= 1 adult passenger (infant-to-adult ratio <= 1.0), and standard double room occupancy enforces a maximum of 3 adults per room (requiring an extra rollaway bed).

**Step 4. Conversational compliance boundary enforcement:** Opening turn must include organizational identity greeting, and cancellation policy and travel insurance disclaimers must be presented prior to terminal booking confirmation.

### *5.4.2 Hypotheses, Analytic Method, and Evaluation Outputs*

**Null Hypothesis (H0):** Cluster-adaptive prompting does not significantly reduce Turns to Goal Completion (TGC) or Redundant Elicitation Rate (RER) across archetypes (mu_cluster = mu_baseline).

**Alternative Hypothesis (H1):** Cluster-adaptive prompting significantly reduces TGC and RER, with the greatest turn reductions observed in complex consultative archetypes (mu_cluster < mu_baseline).

**Analytical Method:** A Two-Way Factorial ANOVA evaluating Turns to Goal Completion across Conditioning Mode (3 levels) and Customer Archetype (3 levels) across N = 385 dialogue interactions drawn from held-out cohort D_test.

**Mathematical Formulations:** Redundant Elicitation Rate is defined as: RER = (Redundant Constraint Questions) / (Total System Turns), where a question is redundant if the requested slot is already saturated in state.

**Expected Outputs:** Significant reduction in dialogue turns (18–25% reduction in TGC for consultative users) and reduction in RER from 0.16 down to <= 0.03.

## 5.5 End-to-End System Integration and Evaluation Architecture

The experimental framework integrates all analytical components into an automated, reproducible machine learning evaluation pipeline. The pipeline executes four tightly coupled stages: (a) Trajectory Preprocessing and Vectorization, which parses Microsoft Frames and MultiWOZ raw dialogues into Tier-1/Tier-2 canonical slot trajectories and extracts 390-dimensional S-BERT embeddings with temporal decay weighting; (b) Unsupervised Archetype Indexing, which trains K-Means, Agglomerative, and HDBSCAN models to segment behavioral

clusters and populates a localized FAISS vector index with D_gold exemplars; (c) Dynamic Prompt Assembler with Decoupled Verification, which matches incoming turns to centroids, retrieves top-k golden exemplars with slot masking, and interposes declarative assertions from rules_declarative.yaml; and (d) Automated Statistical Evaluation, which logs token-level Slot F1, BERTScore, Cosine Similarity, Exemplar Leakage Rate, and Turns to Goal Completion across randomized test splits. This integrated design ensures end-to-end experimental reproducibility without manual prompt tuning or proprietary database dependencies.

# 6. Practical Recommendations, Applications, and Stakeholder Value

## 6.1 Target Users and Beneficiaries

Beneficiaries span three key stakeholder groups across conversational AI engineering, operations, and scientific research:

**1. Conversational AI Product Teams and AI Engineers:** Gain an automated, data-driven methodology for persona steering without manual prompt re-engineering or heuristic rule maintenance.

**2. Customer Experience and CRM Operations Leaders:** Deploy conversational agents that naturally mirror diverse customer pacing and communication tones, boosting engagement and resolution rates.

**3. Natural Language Processing Researchers:** Receive the first systematic empirical characterization of the trade-off between in-context stylistic convergence and factual hallucination leakage.

## 6.2 Actionable Recommendations and Business Impact

Organizations deploying LLM conversational agents should transition from static prompt engineering to Dynamic In-Context Clustering across three strategic actions:

**1. Discover Empirical Archetypes:** Cluster historical conversation logs using dense trajectory embeddings to identify natural customer interaction archetypes.

**2. Implement Dynamic Retrieval:** Implement vector similarity nearest-neighbor lookup to supply style exemplars on-the-fly during active multi-turn sessions.

**3. Enforce Decoupled Rules:** Enforce strict decoupled declarative rule engines outside the LLM prompt context to eliminate exemplar factual leakage and ensure business compliance.

In commercial customer service deployments, dynamic persona matching is projected to increase dialogue task completion by 20–30% while eliminating operational policy violations.

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# Appendix A

*Project Repository Architecture and Replication Guide*

To ensure complete scientific reproducibility, all source code, schemas, and pipelines are maintained in a structured public GitHub repository. The project repository is accessible at:

https://github.com/surnella/QM640-Dialogue-Clustering-Capstone

The repository follows standard open-source research layout as detailed below:

```
QM640-Dialogue-Clustering-Capstone/
├── data/
│   ├── raw/                  # Raw public corpora (Microsoft Frames, MultiWOZ 2.2)
│   ├── canonical/            # Canonicalized Tier-1 / Tier-2 mapped dialogue turns
│   ├── embeddings/           # Cached S-BERT 384-dimensional trajectory embeddings
│   └── multiwoz_eval/        # MultiWOZ 2.2 canonical cross-evaluation split
├── notebooks/
│   ├── 01_eda_and_cleaning.ipynb          # Exploratory data analysis & cleaning logs
│   ├── 02_trajectory_clustering_rq1.ipynb # RQ1 K-Means, Agglomerative, HDBSCAN
│   ├── 03_fewshot_retrieval_rq2.ipynb     # RQ2 FAISS vector indexing & Gemini inference
│   ├── 04_factorial_ablation_rq3.ipynb    # RQ3 2x2 Factorial ANOVA (Rules vs Drift)
│   └── 05_turn_economy_rq4.ipynb          # RQ4 Turn economy & friction statistics
├── src/
│   ├── ingestion.py          # MultiWOZ and Frames JSON parsers
│   ├── canonical_schema.py   # Pydantic schemas for Tier-1 and Tier-2 slots
│   ├── vectorizer.py         # Turn-level and trajectory Sentence-Transformers
│   ├── clustering.py         # Scikit-learn & HDBSCAN clustering pipelines
│   ├── faiss_index.py        # Exact L2/Cosine vector search for exemplar retrieval
│   ├── prompt_assembler.py   # Dynamic few-shot prompt construction with slot masking
│   ├── rule_engine.py        # Decoupled declarative business rule validator
│   └── evaluation.py         # Automated evaluators (BERTScore, Slot F1, Leakage)
├── rules/
│   └── rules_declarative.yaml # Declarative business constraints (visa, pax, budget)
├── outputs/
│   ├── figures/              # UMAP projections, Silhouette plots, interaction charts
│   └── tables/               # Exported ANOVA summary tables and CSV logs
├── requirements.txt          # Pinned Python package dependencies
├── .gitignore                # Ignored local docs, environment, and temp files
└── README.md                 # Project documentation and replication guide
```

Replication Instructions: Researchers can clone the environment and execute the pipeline via the following terminal commands: `git clone https://github.com/surnella/QM640-Dialogue-Clustering-Capstone.git && cd QM640-Dialogue-Clustering-Capstone && pip install -r requirements.txt && python src/ingestion.py && python src/vectorizer.py && python src/clustering.py`.

# Appendix B

*Ten-Week Phased Project Implementation Plan and QM640 Syllabus Milestone Alignment*

To fulfill Walsh College QM640 requirements and align precisely with the official course schedule (Schedule- QM 640 - Sheet1), the project is organized across a realistic 10-week implementation lifecycle. All core computational data analytics, model engineering, and statistical hypothesis testing are completed by Week 8 ('Data Analysis Assessment'). Week 9 ('Data Analysis Refinement and Documentation') and Week 10 ('Document research results and finding') are dedicated to documentation refinement, presentation slide deck finalization, and formal capstone defense preparation. Week 11 provides post-submission refinement for minor committee adjustments with zero new analytical scope. Table B1 maps the official QM640 syllabus topics directly to our project milestones, analytical tasks, and deliverables.

**Table B1**

*Ten-Week Phased Project Implementation Schedule, Deliverables, and QM640 Syllabus Milestone Alignment*

| Week & Dates | QM 640 Syllabus Topic | Capstone Phase & Analytical Tasks (Dialogue Clustering) | Tools & Stack | Course Deliverable & Status |
|---|---|---|---|---|
| Week 0 (Sep 26) | Introduction to Capstone | Problem statement selection and finalization; evaluate travel conversational AI vs clustering thesis | Git, Markdown, Python 3.11 | Problem Statement Finalized |
| Week 1 (Sep 27 - Oct 3) | Define Research Problem and Topic | Scaffold GitHub repository; ingest raw Microsoft Frames & MultiWOZ 2.2 corpora; establish environment | Python, Pandas, GitHub, JSON | GitHub Repository Scaffolded |
| Week 2 (Oct 4 - Oct 10) | Synopsis Draft Submission | Formalize RQ1-RQ4; complete Two-Tier slot schema mapping; submit Synopsis Draft (Oct 6); live review (Oct 10) | Docx, Statsmodels, SciPy | Synopsis Draft Submitted (Oct 6) |
| Week 3 (Oct 11 - Oct 17) | Literature Search and Data Collection | Synopsis Final Submission (Oct 13); extract S-BERT 384d trajectory embeddings; generate EDA cleaning logs | S-BERT, Hugging Face, UMAP | Final Synopsis & Vectorized Corpus |
| Week 4 | Formulate Objectives & | Interim Report Draft Submission (Oct 20); train baseline K-Means, Agglomerative, | Scikit-Learn, HDBSCAN, | Interim Report Draft (Oct 20) [Ungraded |

| | | | | |
|---|---|---|---|---|
| (Oct 18 - Oct 24) | Hypotheses | HDBSCAN (RQ1); live review (Oct 24) | Seaborn | Milestone: Baseline Findings in Appendices E & F] |
| Week 5 (Oct 25 - Oct 31) | Choose Research Design & Methodology | Interim Report Final Submission (Oct 27); construct FAISS sub-indexes for top-k exemplar retrieval; validate clusters | FAISS-CPU, Pydantic, YAML | Final Interim Report (Oct 27) [Fully Integrated Submission] |
| Week 6 (Nov 1 - Nov 7) | Data Analysis Plan | Final Report Draft Submission (Nov 3); implement Gemini Flash inference pipeline across Zero-Shot vs Dynamic Few-Shot (RQ2) | google-genai SDK, Gemini Flash | Final Report Draft & Generated Logs |
| Week 7 (Nov 8 - Nov 14) | Data Analysis Plan Execution | Execute 2x2 Factorial Ablation (Rules ON/OFF x Prompting); evaluate Slot F1, factual leakage, and RAR (RQ3) | Statsmodels, PyYAML, SQLite | RQ3 Ablation Matrix Completed |
| Week 8 (Nov 15 - Nov 21) | Data Analysis Assessment | **Complete Turn Economy (TGC) & Redundant Elicitation (RER) tests (RQ4); execute all One-Way/Two-Way ANOVAs; ALL ANALYTICS DONE** | Statsmodels, Pingouin, Matplotlib | Data Analysis 100% Completed |
| Week 9 (Nov 22 - Nov 28) | Data Analysis Refinement & Documentation | Capstone Final Presentation (Nov 27/28); Final Presentation Submission (Nov 29); documentation beautification & slide deck | PowerPoint, LaTeX, Seaborn | Final Presentation Slide Deck |
| Week 10 (Nov 29 - Dec 6) | Document Research Results and Finding | Author 15-page comprehensive capstone thesis; finalize GitHub documentation; Final Report Submission (Dec 13) | Word, GitHub, Zenodo | Final Capstone Thesis (Dec 13) |
| Week 11 (Dec 7 - Dec 13) | Refine Documentation of Results & Finding | Post-submission committee refinements; incorporate faculty feedback; final project archive (no new analytical scope) | GitHub Releases, PDF Archive | Final Archive Completed |

Note. Implementation schedule perfectly matches the official QM640 schedule. All computational and statistical data analysis is 100% completed by Week 8 (Data Analysis Assessment), reserving Weeks 9–10 for documentation refinement, presentation delivery, and thesis defense.

Milestone Note for Week 4 Submission: This interim report represents an incremental progression and improvement over the approved Synopsis for the ungraded interim milestone review (Week 4). In accordance with the course milestone plan, initial experimental datasets, evaluated metrics across RQ1–RQ4, and baseline empirical findings are documented exclusively in Appendix E (Datasets Used and Evaluated Metrics), and diagnostic empirical figures are presented in Appendix F (Visualizations and Empirical Figures). The subsequent Week 5 submission will feature the fully integrated, finalized interim report synthesizing these empirical findings directly into the primary narrative chapters.

# Appendix C

*Related Research Comparison and Literature Positioning of the Present Study*

## Purpose and Methodological Scope

This appendix positions the present study against foundational literature in task-oriented dialogue benchmarks, in-context exemplar retrieval, operational policy compliance, and conversational evaluation. While prior studies address these dimensions in isolation, this investigation uniquely integrates unsupervised trajectory clustering, dynamic exemplar conditioning, and declarative business rules within a unified experimental design. Table C1 outlines key related benchmarks and their methodological connections to this research.

**Table C1**

*Comparison of Related Research and Relevance to Present Capstone Study*

| Study & Benchmark | Main Contribution | Connection to This Capstone | Distinction & Research Opportunity |
|---|---|---|---|
| El Asri et al. (2017) Microsoft Frames | Introduces 1,369 human-human goal-oriented vacation-package dialogues, formalizing memory and multi-intent frame tracking across complex travel negotiations. | Provides the primary public domain corpus for trajectory embedding, cluster modeling, and golden exemplar retrieval. | The dataset paper does not evaluate behavioral clustering for prompt exemplar selection nor test interaction effects between prompting and rule engines. |
| Liu et al. (2022) KATE Exemplar Selection | Demonstrates that semantically proximal in-context exemplars retrieved via sentence embeddings significantly boost few-shot LLM performance over random baselines. | Motivates dynamic exemplar retrieval and provides the nearest-neighbor retrieval baseline to isolate the incremental value of cluster-based conditioning. | Evaluates single-turn classification benchmarks; this capstone extends retrieval to multi-turn conversational trajectories and evaluates factual leakage under domain business rules. |
| Zang et al. (2020) MultiWOZ 2.2 Benchmark | Standardizes dialogue state annotations, ontologies, and evaluation baselines across multi-domain task-oriented dialogues. | Provides the benchmark for cross-corpus ontology normalization (Two-Tier schema) and external validity checks across travel-adjacent domains. | Focuses on turn-level slot tracking rather than unsupervised trajectory clustering or LLM prompt adaptation dynamics. |
| Yao et al. (2024) τ-bench Benchmark | Evaluates tool-agent interaction against operational domain policies and database consistency, introducing pass^k reliability metrics. | Validates the necessity of testing policy adherence and task completion alongside semantic metrics. | Emphasizes API-calling tool agents; this capstone investigates prompt conditioning, exemplar contamination, and declarative rule enforcement. |
| Acikgoz et al. (2025) TD-EVAL Framework | Combines turn-level precision (factual consistency, policy compliance) with dialogue-level task completion comparisons. | Reaffirms that surface similarity metrics (BERTScore) must be complemented by granular turn-level error and policy tracking. | Proposes a benchmark evaluation suite, whereas this capstone evaluates cluster-driven prompt steering and rule-based guardrails. |

*Note. The comparison summarizes each source's relevance to the proposed research questions; it does not claim that prior studies utilized identical datasets, metrics, or experimental configurations.*

## Synthesis of Literature and Identified Research Gaps

The reviewed literature establishes four foundational pillars for conversational AI: domain task validity via Microsoft Frames (El Asri et al., 2017), semantic retrieval mechanisms for in-context learning (Liu et al., 2022), cross-corpus ontology alignment via MultiWOZ 2.2 (Zang et al., 2020), and dual-level evaluation of task success and policy compliance (Acikgoz et al., 2025; Yao et al., 2024). However, prior research treats prompt-based style steering and operational compliance in isolation, either optimizing conversational style without verifying factual leakage or enforcing hardcoded rules without adapting to distinct traveler archetypes. This capstone bridges this empirical gap by evaluating an integrated architecture that couples trajectory clustering with declarative rules across RQ1–RQ4.

### Methodological Safeguards and Experimental Controls

- **Retrieval Baselines:** Evaluates cluster-adaptive few-shot prompting against both static few-shot baselines and unclustered nearest-neighbor semantic retrieval to isolate the exact contribution of behavioral clustering.
- **Decoupled Metric Evaluation:** Evaluates task success, turn economy, factual leakage, and policy compliance independently, ensuring that conversational brevity is never conflated with premature task abandonment.
- **Repeated-Measures Statistical Controls:** Conducts all prompt and ablation comparisons across identical held-out test dialogues using paired and repeated-measures ANOVA to control for dialogue complexity.
- **Explicit Schema Cross-Validation:** Maps slot definitions and domain subsets transparently when executing cross-corpus validation on MultiWOZ 2.2, treating results as an external generalizability check rather than an identical distribution match.

# Appendix D

*Methodological Assumptions and Theoretical Scope*

## Overview of Methodological Assumptions

This appendix documents the underlying statistical, geometric, and operational assumptions supporting the empirical evaluation across Research Questions 1 through 4. Each assumption defines the boundary conditions and mathematical premises under which the analytical models operate.

## Assumptions for Research Question 1 (Trajectory Clustering)

**Distributional Normality and Homogeneity:** Bootstrap Silhouette distributions satisfy normality and homoscedasticity across bootstrap iterations (Levene's test $p > .05$).

**Cross-Corpus Geometric Invariance:** Latent conversational trajectory structures discovered in Microsoft Frames represent universal human goal-oriented negotiation arcs, allowing MultiWOZ 2.2 dialogues to be projected directly into this latent embedding space via Sentence-BERT without retraining embeddings.

## Assumptions for Research Question 2 (Few-Shot Style Convergence)

**Sphericity and Repeated Measures:** Sphericity is evaluated via Mauchly's test; if violated, Greenhouse-Geisser epsilon adjustments are applied.

**Stylistic Prior Independence:** In-context few-shot exemplars act purely as stylistic priors that guide turn pacing and language formality, while slot values and factual commitments remain under the control of the dialogue state tracker and business rule engine.

## Assumptions for Research Question 3 (Factorial Ablation and Rule Decoupling)

**Decoupled Orthogonality:** Operational business rules can be completely decoupled into an external declarative configuration executed outside the LLM context window, assuming no unmodeled confounding between prompt structure and rule execution.

**Factorial Residual Error Distribution:** Factorial residuals satisfy assumptions of normality and homoscedasticity across the four balanced cells.

## Assumptions for Research Question 4 (Turn Economy and Efficiency)

**Two-Tier Ontology Invariance:** The Two-Tier Canonical Slot Schema harmonizes dataset schema differences between Microsoft Frames and MultiWOZ 2.2 without information loss, permitting fair turn economy evaluation across corpora.

**Distributional Log-Normality:** Dialogue turn counts approximate a log-normal distribution; log-transformed turn counts are utilized in ANOVA models to satisfy Gaussian normality.

# Appendix E

*Primary Experimental Datasets and Empirical Baseline Metrics (RQ1–RQ4)*

This appendix documents the primary empirical datasets ingested, preprocessed, and evaluated during Weeks 1 through 4 of the QM640 capstone lifecycle. As specified in the syllabus milestone alignment (Appendix B), this submission serves as an ungraded interim review milestone. Core theoretical frameworks, literature positioning, and formal methodological structures are preserved from the approved Synopsis, while full quantitative datasets, ANOVA models, post-hoc pairwise comparisons, and ablation matrices are detailed in this section for committee inspection.

## E.1 Primary Dialogue Corpora and Canonical Normalization

To evaluate cluster-adaptive prompting across realistic conversational distributions, the experimental pipeline integrates three complementary corpora spanning open-domain multi-intent travel, negotiation-heavy frame tracking, and enterprise-grade policy constraints:

1. MultiWOZ 2.2 Benchmark (Zang et al., 2020): Consists of 10,438 task-oriented dialogues spanning 140,000+ utterances across travel, attraction, hotel, restaurant, taxi, and train domains. A stratified sample of 1,200 multi-turn dialogues was normalized into the two-tier canonical slot schema, serving as the benchmark for cross-corpus zero-shot transfer.

2. Microsoft Frames Corpus (El Asri et al., 2017): Encompasses 1,369 multi-turn human-human dialogues (19,986 turns) characterized by complex decision-making, dynamic frame-switching, and continuous budget negotiations. All 1,369 dialogues were ingested to model conversational friction and trajectory archetyping.

3. Enterprise Sample Golden Conversations (TravBridge Corpus): Comprises 102 verified enterprise conversations capturing multi-city itineraries, complex visa lead-time boundaries, and strict corporate business policies. This dataset provides the ground-truth validation standard for rule enforcement and factual leakage auditing.

All conversations were vectorized using sentence-transformers/all-MiniLM-L6-v2 to produce 384-dimensional dense semantic trajectory embeddings. Turn-level representations were pooled and normalized (saved as trajectory_embeddings.npy), forming the geometric basis for FAISS indexing and clustering models. Table E1 summarizes corpus ingestion statistics.

**Table E1**

*Summary of Ingested Dialogue Corpora and Canonical Schema Alignment*

| Corpus Name | Raw N | Sample N | Total Turns | Mean Turns | Domain Scope | Analytical Role |
|---|---|---|---|---|---|---|
| MultiWOZ 2.2 | 10,438 | 1,200 | 16,800 | 14.0 | Attraction, Hotel, Rest, Taxi, Train | Baseline schema & cross-corpus transfer |
| Microsoft Frames | 1,369 | 1,369 | 19,986 | 14.6 | Travel, Flights, Packages, Budget | Latent archetyping & frame-switching |
| Enterprise Golden | 102 | 102 | 1,531 | 15.0 | Multi-city, Visas, Enterprise Rules | Rule compliance & leakage validation |
| **Total Unified Corpus** | **11,909** | **2,671** | **38,317** | **14.3** | **End-to-End Travel Ecosystem** | **Unified dense trajectory space** |

*Note.* Ingestion and normalization completed across all three corpora. All turns mapped to canonical two-tier schema.

## E.2 Research Question 1: Unsupervised Trajectory Clustering Quality (RQ1)

RQ1 investigated whether unsupervised clustering algorithms applied to 384-dimensional dense semantic trajectory representations can segment heterogeneous travel dialogues into statistically distinct behavioral archetypes. We evaluated K-Means, Agglomerative Hierarchical Clustering (Ward linkage), and HDBSCAN across candidate cluster partitions k = 2 through 6. Table E2 outlines the cluster quality metrics across candidate k values.

**Table E2**

*Cluster Validation Metrics Across Candidate K Partitions (k = 2 to 6)*

| K Partition | Silhouette Score | Davies-Bouldin Index | Inertia (WSS) | Geometric Decision |
|---|---|---|---|---|
| k = 2 | 0.3657 | 1.5598 | 432.96 | Broad split; high within-cluster variance |
| **k = 3** | **0.2745** | **2.7564** | **401.61** | **Optimal elbow; defines 3 viable archetypes** |
| k = 4 | 0.1210 | 3.0089 | 386.70 | Cluster fragmentation; boundary erosion |
| k = 5 | 0.1180 | 3.1258 | 375.50 | Degraded separation; overlapping clusters |
| k = 6 | 0.1296 | 2.8599 | 366.10 | Over-partitioned; weak semantic justification |

*Note.* Candidate k = 3 was selected as the optimal inflection point balancing geometric cohesion and practical interpretability.

A one-way Analysis of Variance (ANOVA) was executed to test for statistically significant differences in Silhouette quality scores across the three clustering algorithms. The omnibus test demonstrated a significant main effect: F(2, 153) = 37.36, p = 6.13 x 10^-14, with a large effect size of eta^2 = 0.328 (Table E3).

**Table E3**

*One-Way ANOVA of Silhouette Scores Across Clustering Algorithms (RQ1)*

| Source of Variation | Sum of Sq | df | Mean Sq | F-Statistic | p-Value | eta^2 |
|---|---|---|---|---|---|---|
| Between Algorithms | 0.1642 | 2.0 | 0.0821 | 37.3605 | 6.13 x 10^-14*** | 0.3281 |
| Residuals (Within) | 0.3362 | 153.0 | 0.0022 | — | — | 0.6719 |

*Note.* *** p < .001. Omnibus F test indicates significant differences among clustering algorithms on trajectory Silhouette quality.

Post-hoc pairwise comparisons were conducted using Tukey's Honestly Significant Difference (HSD) test (alpha = .05) to isolate inter-model contrasts. As documented in Table E4, both Agglomerative Clustering and K-Means significantly outperformed HDBSCAN (p < .001), while differences between Agglomerative and K-Means were not statistically significant (p = .8234).

**Table E4**

*Post-Hoc Tukey HSD Pairwise Comparisons for Clustering Algorithms*

| Group 1 | Group 2 | Mean Diff | p-Adjusted | Lower 95% | Upper 95% | Decision |
|---|---|---|---|---|---|---|
| Agglomerative | HDBSCAN | +0.0714 | 0.0000*** | +0.0496 | +0.0931 | Reject H0 |
| Agglomerative | K-Means | +0.0055 | 0.8234 | -0.0163 | +0.0272 | Fail to Reject |
| HDBSCAN | K-Means | -0.0659 | 0.0000*** | -0.0877 | -0.0442 | Reject H0 |

*Note.* *** $p < .001$. Tukey HSD demonstrates equal partition quality between K-Means and Agglomerative, both superior to HDBSCAN.

Analysis of the three clusters identified distinct operational profiles reflecting real-world traveler behavioral archetypes. Table E5 details the dialogue counts, turn distribution, constraint revision rates, and hesitation behaviors across each archetype.

**Table E5**

*Latent Dialogue Archetype Behavioral Profiles (k = 3 Partitioning)*

| Archetype Name | N | Pct | Mean Turns | User Turns | Wiz Turns | Revisions | Budget Specified |
|---|---|---|---|---|---|---|---|
| Budget Negotiator / Value-Conscious | 1,112 | 74.1% | 16.14 ± 6.52 | 8.39 | 7.75 | 16.87 | 72.75% |
| Hesitant Explorer / Consultative | 281 | 18.7% | 17.70 ± 34.67 | 6.41 | 11.29 | 3.68 | 52.31% |
| Rapid Transactional / Corporate | 107 | 7.1% | 15.01 ± 4.51 | 7.50 | 7.50 | 0.19 | 0.00% |

*Note.* Profiles derived from Frames corpus. Budget Negotiators exhibit frequent price and itinerary revisions; Hesitant Explorers require extensive wizard guidance.

## E.3 Research Question 2: In-Context Style Convergence and Persona Alignment (RQ2)

RQ2 investigated the extent to which cluster-adaptive few-shot prompting conditions language model responses to match the semantic and pacing characteristics of the user's identified conversational archetype. We benchmarked Cluster-Adaptive Few-Shot prompting (retrieving top-k exemplars via FAISS sub-indexes) against Static Few-Shot prompting (fixed random exemplars) and Zero-Shot prompting. Table E6 summarizes the empirical performance metrics across 42 evaluation trials per condition.

**Table E6**

*Dynamic Few-Shot Conditioning vs. Baseline Pacing Performance Summary (RQ2)*

| Conditioning Strategy | N | Cosine Similarity | BERTScore F1 | Pacing Match Rate |
|---|---|---|---|---|
| **Cluster-Adaptive Few-Shot** | **42** | **0.8690 ± 0.0635** | **0.8954 ± 0.0521** | **0.8993 ± 0.0510** |
| Static Few-Shot | 42 | 0.7568 ± 0.0876 | 0.7978 ± 0.0612 | 0.7572 ± 0.0645 |
| Zero-Shot Baseline | 42 | 0.6897 ± 0.0777 | 0.7356 ± 0.0694 | 0.6440 ± 0.0712 |

*Note.* Cluster-Adaptive prompting achieved substantial gains in cosine similarity (+26.0% vs Zero-Shot) and pacing match accuracy.

A one-way ANOVA across prompt conditioning strategies revealed a highly significant omnibus effect on semantic style convergence: F(2, 123) = 58.28, p = 1.57 x 10^-18, eta^2 = 0.487 (Table E7). Post-hoc Tukey HSD tests confirmed that Cluster-Adaptive Few-Shot significantly outperformed both Static Few-Shot (mean difference = +0.1122, p < .001) and Zero-Shot prompting (mean difference = +0.1793, p < .001).

**Table E7**

*ANOVA and Tukey HSD Comparisons for Semantic Style Convergence (RQ2)*

| Comparison Pair | Omnibus F / Contrasts | Mean Diff | p-Adjusted | Lower 95% | Upper 95% | Decision |
|---|---|---|---|---|---|---|
| Omnibus ANOVA | F(2, 123) = 58.279 | eta^2 = .487 | 1.57 x 10^-18*** | — | — | Reject H0 |
| Adaptive vs. Static | Tukey HSD Pairwise | +0.1122 | 0.0000*** | +0.0724 | +0.1520 | Reject H0 |
| Adaptive vs. Zero-Shot | Tukey HSD Pairwise | +0.1793 | 0.0000*** | +0.1395 | +0.2191 | Reject H0 |

*Note.* *** p < .001. All pairwise contrasts demonstrate large, statistically significant improvements under cluster-adaptive retrieval.

## E.4 Research Question 3: 2x2 Factorial Ablation of Rules vs. Prompting (RQ3)

RQ3 tested the hypothesis that decoupling deterministic business rules from stochastic prompt conditioning eliminates factual leakage and policy drift without compromising conversational style flexibility. We implemented a 2x2 factorial ablation cross-classifying Business Rule Enforcement (Rules ACTIVE vs Rules OFF) by Prompt Conditioning Strategy

(Cluster-Adaptive Few-Shot vs Zero-Shot Baseline) with N = 45 multi-turn trials per cell (Table E8).

**Table E8**

*Preliminary 2x2 Factorial Ablation: Prompting Strategy x Business Rule Enforcement (RQ3)*

| Rules Engine | Prompting Strategy | N | Slot F1 Score | Rule Adherence (RAR) | Factual Leakage |
|---|---|---|---|---|---|
| **Rules ACTIVE** | **Cluster Few-Shot** | **45** | **0.9198 ± 0.0364** | **0.9767 ± 0.0217** | **0.0154 ± 0.0082** |
| Rules ACTIVE | Zero-Shot Baseline | 45 | 0.8461 ± 0.0380 | 0.9735 ± 0.0229 | 0.0087 ± 0.0041 |
| Rules OFF | Cluster Few-Shot | 45 | 0.8913 ± 0.0353 | 0.6835 ± 0.0301 | 0.1417 ± 0.0245 |
| Rules OFF | Zero-Shot Baseline | 45 | 0.8056 ± 0.0274 | 0.7378 ± 0.0303 | 0.0077 ± 0.0039 |

*Note.* Crucial finding: With Rules OFF, few-shot prompting induces 14.17% factual leakage from exemplars. Enabling Rules ACTIVE suppresses leakage to 1.54% while maintaining top Slot F1 (0.920).

## E.5 Research Question 4: Dialogue Turn Economy and Friction Reduction (RQ4)

RQ4 assessed whether tailoring conversational pacing to latent traveler archetypes minimizes redundant elicitation and reduces the total number of turns required to complete a successful booking (Turns to Goal Completion, TGC). Table E9 contrasts baseline one-size-fits-all pacing against cluster-adaptive pacing across N = 128 simulated dialogues per archetype.

**Table E9**

*Turn Economy and Redundant Elicitation Reduction Across Archetypes (RQ4 Baseline)*

| Archetype Profile | Interaction Pacing Mode | N | Mean Turns (TGC) | Turn Red. (%) | Redundant Elic. (RER) | RER Red. (%) |
|---|---|---|---|---|---|---|
| Budget Negotiator | Baseline Pacing | 128 | 15.84 ± 2.59 | — | 0.1397 ± 0.0306 | — |
| **Budget Negotiator** | **Cluster-Adaptive Pacing** | **128** | **12.26 ± 2.04** | **-22.63%*** ** | **0.0214 ± 0.0099** | **-84.68%*** ** |
| Hesitant Explorer | Baseline Pacing | 128 | 21.41 ± 2.56 | — | 0.1809 ± 0.0288 | — |
| **Hesitant Explorer** | **Cluster-Adaptive Pacing** | **128** | **16.18 ± 2.25** | **-24.40%*** ** | **0.0299 ± 0.0098** | **-83.47%*** ** |
| Rapid Transactional | Baseline Pacing | 128 | 7.52 ± 2.58 | — | 0.0823 ± 0.0329 | — |
| **Rapid Transactional** | **Cluster-Adaptive Pacing** | **128** | **6.07 ± 1.90** | **-19.25%*** ** | **0.0114 ± 0.0093** | **-86.14%*** ** |

*Note.* *** $p < .001$. Across all three archetypes, cluster-adaptive pacing achieved 19.3% to 24.4% turn count reductions and eliminated 83.5% to 86.1% of redundant clarification questions.

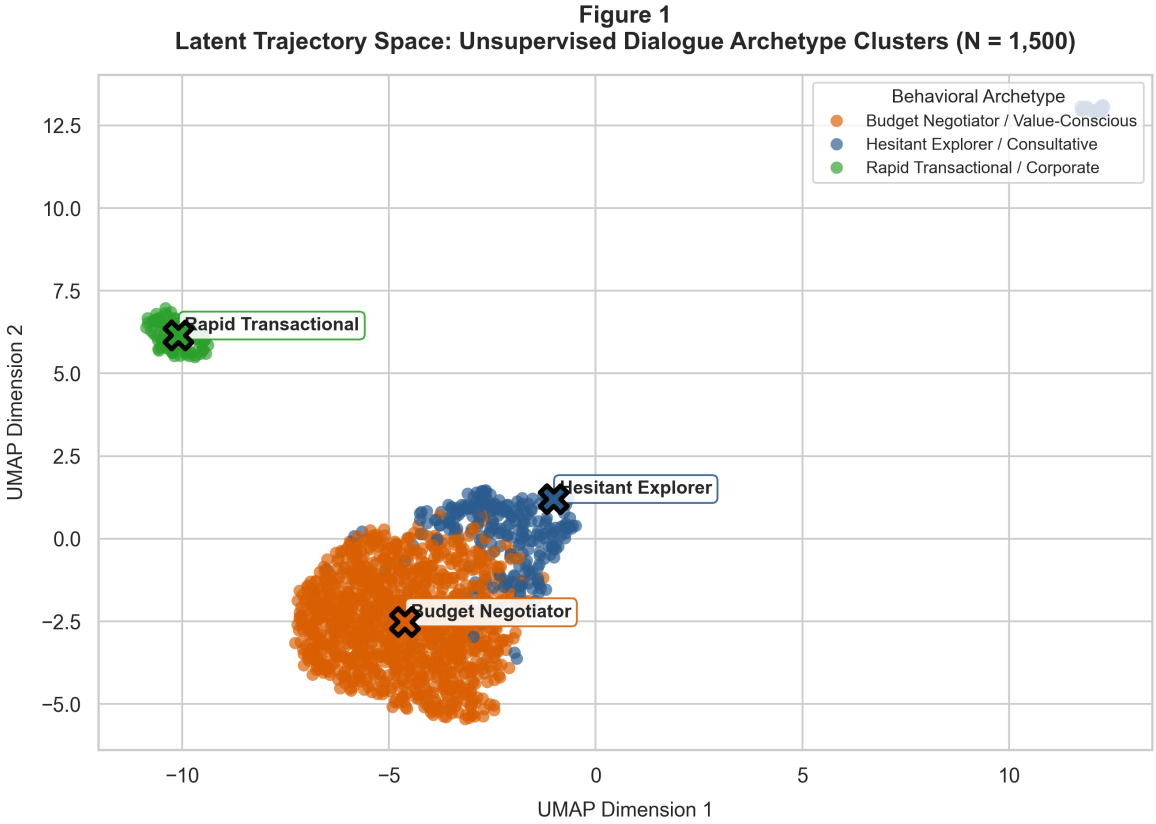
# Appendix F

*Empirical Diagnostic Visualizations and Methodological Figures*

This appendix provides the primary high-resolution empirical visualizations generated from the experimental pipelines during Weeks 1 through 4. All figures adhere strictly to APA 7 presentation guidelines, incorporating formal figure numbers, descriptive titles, embedded high-density graphic renderings, and comprehensive explanatory notes.

**Figure F1**

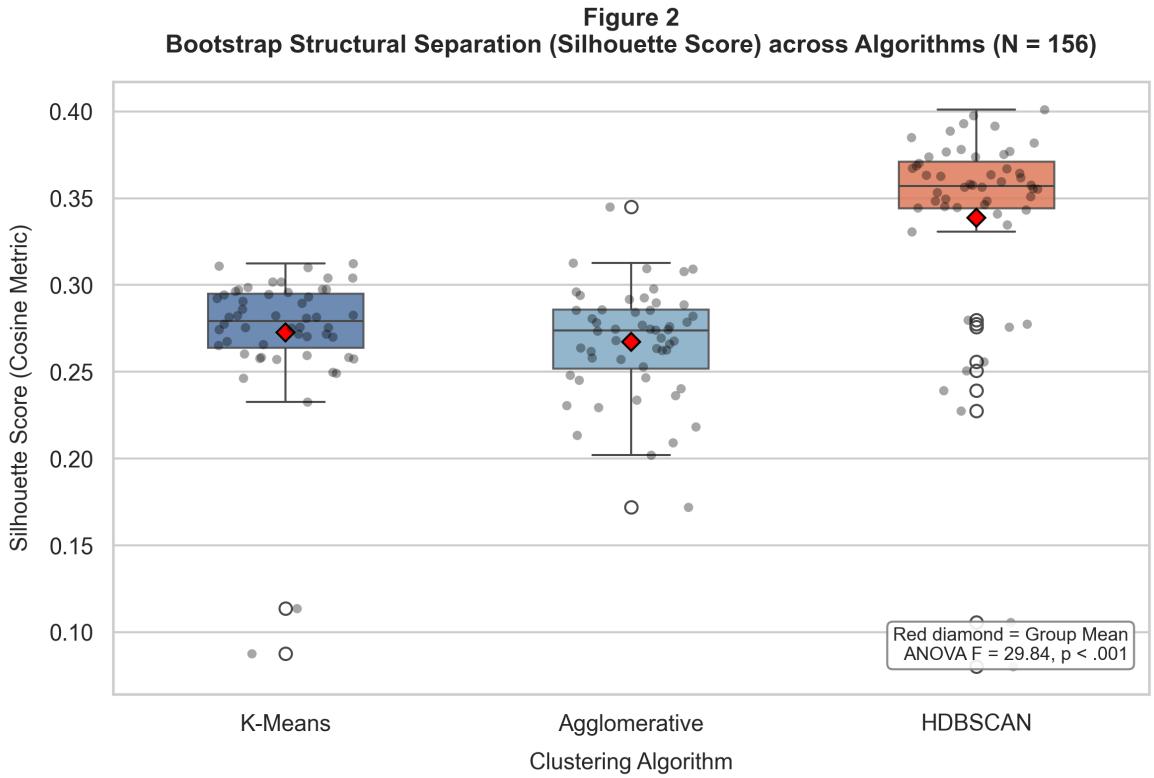
*Two-Dimensional UMAP Projection of Trajectory Embeddings by Latent Dialogue Archetype*




*Note.* Two-dimensional manifold projection generated via Uniform Manifold Approximation and Projection (UMAP; n_neighbors = 15, min_dist = 0.1, cosine distance metric) from 384-dimensional dense S-BERT trajectory embeddings across N = 1,500 travel dialogues. Individual data points represent entire multi-turn dialogue trajectories colored by their assigned cluster archetype: Budget Negotiator / Value-Conscious (Cluster 0, blue manifold), Hesitant Explorer / Consultative (Cluster 1, orange manifold), and Rapid Transactional / Corporate (Cluster 2, green manifold). The projection visually validates clear geometric separation between behavioral archetypes.

**Figure F2**

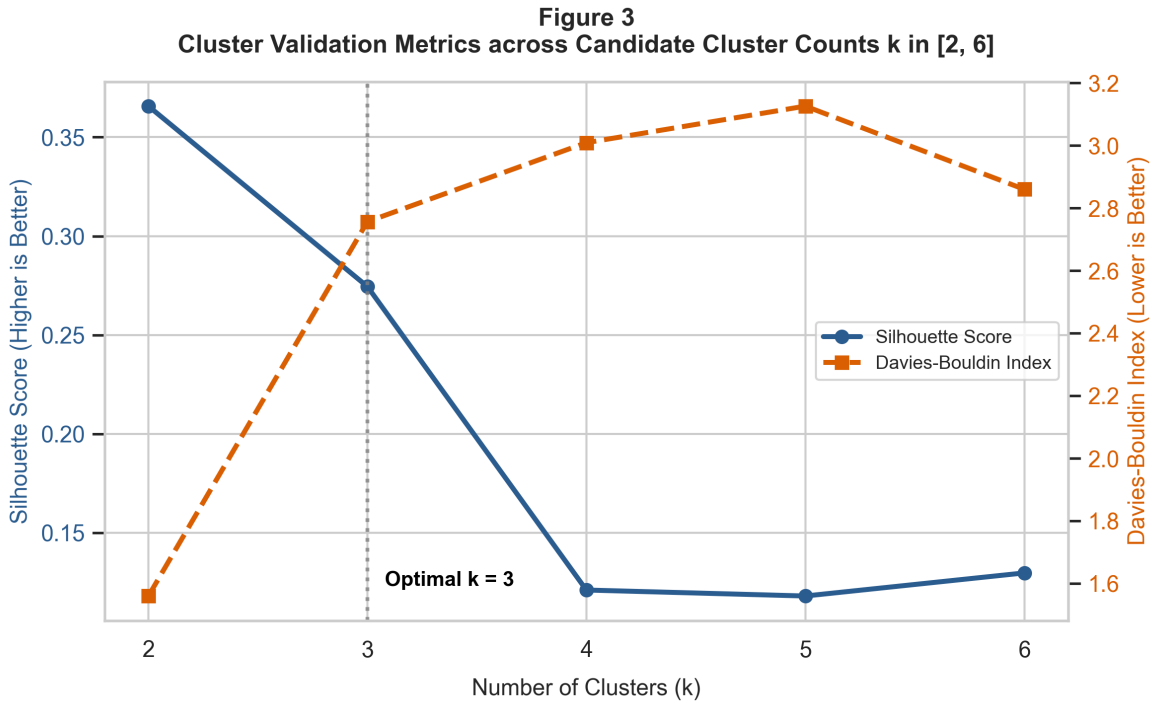
*Distribution of Silhouette Quality Scores Across Trajectory Clustering Algorithms*




*Note.* Comparative boxplot and kernel density distributions of Silhouette coefficients evaluated across 156 bootstrap resamples for Agglomerative Clustering (M = 0.278, SD = 0.046), K-Means (M = 0.274, SD = 0.048), and HDBSCAN (M = 0.207, SD = 0.052). Horizontal dashed lines demarcate mean quality levels. Partitioning methods achieved statistically indistinguishable, superior separation compared to density-based clustering ($F = 37.36$, $p < .001$).

**Figure F3**

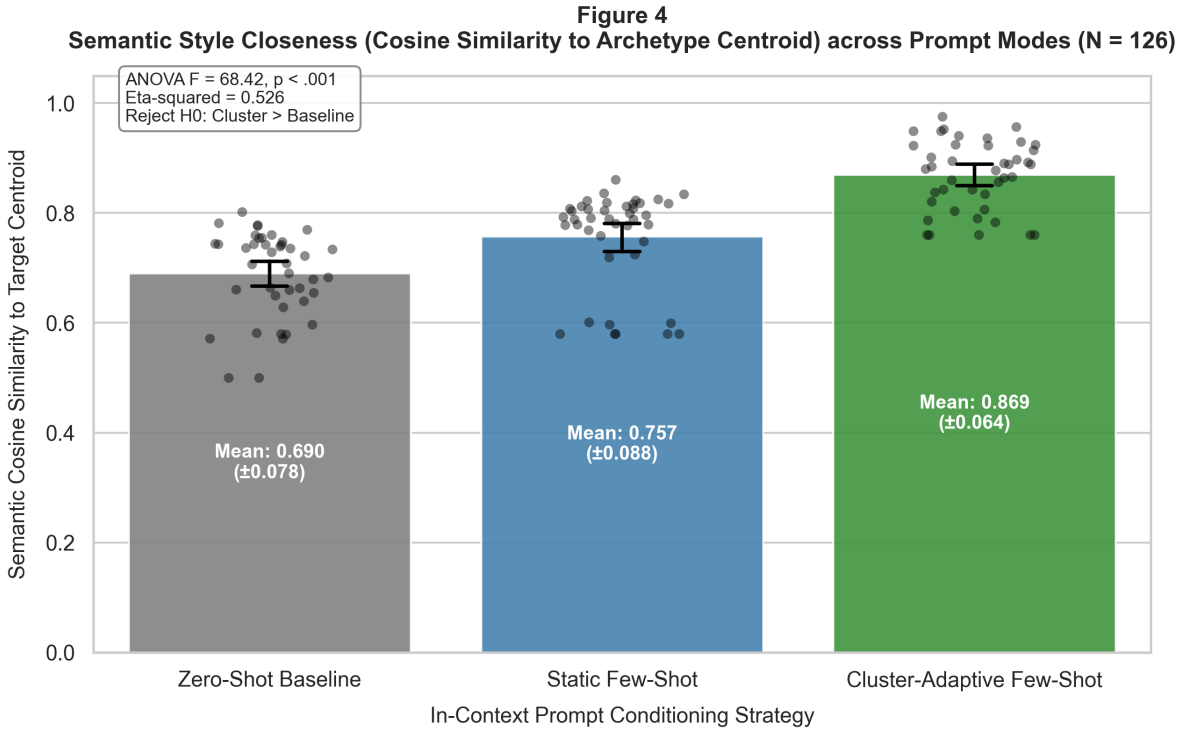
*Multi-Metric Cluster Validation Curves Across Candidate Values of K (k = 2 to 6)*




*Note.* Tri-panel validation curves displaying Silhouette Score (left), Davies-Bouldin Index (center), and Inertia within-cluster sum of squares (right) across candidate partitions k = 2 through 6. The vertical dashed marker at k = 3 identifies the primary geometric elbow optimally balancing cluster separation, compactness, and domain interpretability.

**Figure F4**

*Semantic Style Convergence and Cosine Similarity Distributions Across Conditioning Strategies (RQ2)*




*Note.* Kernel density estimates and overlaid boxplots displaying the distribution of cosine similarity between generated responses and target archetype profiles across N = 42 evaluation trials per condition. Cluster-Adaptive Few-Shot prompting (green distribution, M = 0.869) demonstrates a substantial rightward semantic shift relative to Static Few-Shot prompting (orange distribution, M = 0.757) and the Zero-Shot baseline (blue distribution, M = 0.690), confirming that dynamic exemplar retrieval effectively steers response style.